

The constant below the logarithm, general equal case:

$$Z(n) = \frac{n^{-p/2}}{k_1 k_2} \left(A_{p-1} \left(\frac{1}{2} \log n + (k_1 + k_2) \log b \right) + C_0(p-1) \right) + O(n^{-(p+1)/2})$$

Claude, with Daniel Murfet

2026-09-06

Abstract

Unit 41's identification of the constant below $\log n$ is carried over to the general equal-exponent case of unit 45. The weighted logarithmic primitive obeys the exact identity $\int_0^L F_\gamma/x - A_\gamma \log L - C_0(\gamma) = \int_L^\infty (A_\gamma - F_\gamma)/x \in [0, A_{\gamma+1}/L]$ with $C_0(\gamma) = \int_0^1 F_\gamma/x - \int_1^\infty (A_\gamma - F_\gamma)/x$, giving the two-term expansion in the title with an explicit remainder and the limit $n^{p/2}Z(n) - \frac{A_{p-1}}{2k_1k_2} \log n \rightarrow \frac{1}{k_1k_2} (A_{p-1}(k_1 + k_2) \log b + C_0(p-1))$. Zero sorry/axiom.

1 Setting

The tail integrand $(A_\gamma - F_\gamma(x))/x$ is nonnegative and at most $A_{\gamma+1}x^{-2}$ by unit 45's tail bound, hence integrable on $[1, \infty)$ with $\int_L^\infty \leq A_{\gamma+1}/L$. The split of unit 45 (`weightedLogPrimitive_split`) then makes the constant term and its one-sided rate immediate, exactly as at $\gamma = 0$.

2 The things formalised

```
noncomputable def weightedLogConst ( a : ) : :=
  ( x in Ioc 0 1, weightedPrimitive a x / x)
  - x in Ioi 1, (weightedMass a - weightedPrimitive a x) / x
theorem weightedLogPrimitive_sub_log_sub_const ... (hL : 1 L) :
  weightedLogPrimitive a L - weightedMass a * Real.log L - weightedLogConst a
  = x in Ioi L, (weightedMass a - weightedPrimitive a x) / x
theorem twoDGeneral_second_order_isBig0 ...
  (hp : ((h : ) + 1) / k = ((h : ) + 1) / k) :
  (fun n => twoDGeneral a b n h h k k
    - n ^ (-(p / 2)) * twoDGeneralSecondOrderCoeff a b p k k n)
  = 0[atTop] fun n => n ^ (-(p / 2) - 1 / 2) -- p = (h+1)/k
theorem twoDGeneral_second_order_tendsto ... :
  Tendsto (fun n => n ^ (p / 2) * twoDGeneral a b n h h k k
    - weightedMass a (p - 1) / (2 * ((k : ) * k)) * Real.log n)
  atTop ( (1 / ((k : ) * k)) * (weightedMass a (p - 1)
    * (((k : ) + k) * Real.log b) + weightedLogConst a (p - 1))))
```

3 Proof strategy

Identical in structure to unit 41 with F, A, M replaced by $F_\gamma, A_\gamma, A_{\gamma+1}$: the $(1, L] \cup (L, \infty)$ split of the tail integral, comparison with $A_{\gamma+1} \int_L^\infty x^{-2}$, and the two-sided bound $0 \leq n^{p/2} Z - \text{coeff}(n) \leq \frac{A_p}{k_1 k_2 b^{k_1+k_2}} n^{-1/2}$ for $n \geq \max(1, b^{-2(k_1+k_2)})$. The big- O uses $n^{-p/2} n^{-1/2} = n^{-p/2-1/2}$ via `Real.rpow_sub`; the limit is `squeeze_zero`.

4 Roadblocks and resolutions

- Rewriting `p - 1 + 1 = p` at two hypotheses fails when one of them has no such subterm; rewrite each hypothesis separately.
- `rw [one_div]` picked the instance `1/2` inside an exponent instead of `1/√n`; after `div_eq_mul_inv`, use `← one_div` to turn the unique inverse into `1/√n`.

5 What was Mathlib, what was new

Mathlib: `integral_Ioi_rpow_of_lt`, `integrableOn_Ioi_rpow_of_lt`, `setIntegral_union`, `Real.rpow_sub`, `squeeze_zero`. New: the weighted constant term and the two-term expansion of the general equal case.

6 Lessons

A parametrised lemma family (unit 45) pays off immediately: the second-order theory that took a dedicated unit at $\gamma = 0$ transfers in one pass once the weighted bounds exist.

7 Outcome

The general two-dimensional equal-exponent integral now has a two-term expansion with explicit constant and $O(n^{-(p+1)/2})$ remainder, completing the two-dimensional programme of units 44–48 (exact reduction, three regimes, trichotomy, next-order term). Remaining directions: mixed multiplicity in general dimension, and the general- (k, h) extension of the d -fold iterated primitives.